



Determination of chemical composition (wt.%) in Plain Carbon Steel through Optical Emission Spectroscopy.



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Introduction:

Optical Emission Spectroscopy is a widely utilized analytical technique in the metal industry for determining the chemical composition of metallic materials, especially plain carbon steel. Though its accuracy is less than XRF, it is valued for its speed and for simultaneously detecting multiple elements in a single sample. In OES, a strong spark excites the atoms in a metal, causing them to emit characteristic light at different wavelengths, which is later analyzed to identify and quantify the elements present in that specimen. Basically, the emitted light is separated into its component wavelengths by a monochromator/diffraction grating. There are two detection systems in OES: Qualitative and Quantitative (intensity). Detectors measure the intensity of each wavelength, which corresponds to the concentration of each element in the sample.

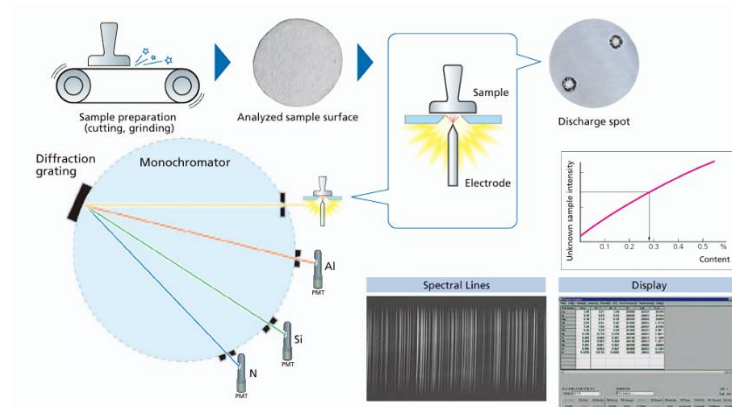


Figure 1: Principle of OES

The primary advantage of OES is that it can also detect small trace elements present in a steel sample if there is a detector for them. It is highly efficient for industry applications because, compared to other techniques, it needs minimal sample preparation and delivers results within minutes. So, in steel production and other metallurgical processes, OES is routinely used for quality assurance, process controls, and compliance with product standards. This experiment focused on determining the chemical composition of plain carbon steel using Optical Emission Spectroscopy (OES) and the best suitable field of our sample using the test result.

Apparatus and Equipment:

1. Plain carbon steel sample (12mm)
2. OES Machine
3. Grinding and Polishing Machine
4. CRM (Certified reference material)

5. Hammer
6. 120 Emery paper
7. Hacksaw and vice

Experimental procedure:

1. First, we collected a 12 mm plain carbon steel sample from our sir.s
2. Then we hammered it to make it flat, because 12 mm is not appropriate to take spark indentation in OES.
3. We then cut a 10 mm (height) piece from the long PCS using a hacksaw and vice.
4. Next, we ground the sample in a grinding machine to make the two surfaces almost flat.
5. After that, we ground it with 120 emery paper to fully flatten the upper and lower surfaces.
6. The sample was immediately taken to the OES lab for testing, before any kind of contamination or oxidation occurred.
7. The OES machine produced sparks on our sample, which excited the atoms at those particular sites. This generated characteristic wavelengths, which were later analyzed by software to detect and quantify the elements.
8. We took measurements at three different spots on our sample and later averaged the results for better accuracy.



Figure 2: Our Sample

Calculation and Result:

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Sample Testing of different Qualities

Chemical Results

Sample ID : GR-2 Material :

Customer : Lab Name : OES

Grade:

Foundry-MASTER Program : FE_100

	Fe	C	Si	Mn	P	S	Cr	Mo
1	97.91	0.196	0.380	0.904	0.0292	0.0516	0.100	0.0072
2	97.98	0.190	0.372	0.890	0.0294	0.0472	0.100	0.0081
3	97.94	0.188	0.382	0.908	0.0297	0.0488	0.100	0.0080
Ave	97.95	0.191	0.378	0.901	0.0294	0.0492	0.100	0.0077
	Ni	Al	Co	Cu	Mg	Nb	Ti	V
1	0.0402	0.0056	0.0060	0.156	0.0022	0.0051	0.0048	0.0072
2	0.0402	0.0040	0.0064	0.153	0.0014	0.0026	0.0050	0.0059
3	0.0407	0.0033	0.0070	0.154	0.0012	0.0196	0.0058	0.0065
Ave	0.0404	0.0043	0.0065	0.154	0.0016	0.0091	0.0052	0.0065
	W	Pb	Sn	B	Ca	Zr	Zn	Bi
1	< 0.0050	< 0.0050	0.0222	0.0023	0.0036	0.0063	> 0.0500	0.0021
2	< 0.0050	< 0.0050	0.0225	0.0020	0.0021	0.0064	0.0480	0.0024
3	0.0053	< 0.0050	0.0224	0.0021	0.0015	0.0064	0.0489	0.0036
Ave	< 0.0050	< 0.0050	0.0224	0.0021	0.0024	0.0064	0.0493	0.0027
	As	N	Se	Sb	Ta			
1	0.0016	0.0797	< 0.0010	< 0.0010	0.0175			
2	0.0021	0.0625	< 0.0010	< 0.0010	0.0169			
3	0.0024	0.0411	< 0.0010	< 0.0010	0.0181			
Ave	0.0020	0.0611	< 0.0010	< 0.0010	0.0175			

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Figure 3: OES test result of our sample

CRM value (elements% in this test)

Elements	Fe	C	Si	Mn	P	S
%elements	97.87	0.286	0.174	0.831	0.0132	0.0282

CRM value (element% in first test)

Elements	Fe	C	Si	Mn	P	S
%elements	97.87	0.297	0.206	0.879	0.0238	0.0371

We know that,

$$\text{Correction factor} = \frac{\text{Element (\% in CRM in first test)}}{\text{Element (\% in CRM in this test)}}$$

Determination of chemical composition in wt.% = Correction Factor $\times$ element (%) in sample

Correction factor for,

$$\text{Fe} = 1;$$

$$\text{C} = \frac{0.297}{0.286} = 1.0385$$

$$\text{Si} = \frac{0.206}{0.174} = 1.1839$$

$$\text{Mn} = \frac{0.879}{0.831} = 1.0578$$

$$\text{P} = \frac{0.0238}{0.0132} = 1.803$$

$$\text{S} = \frac{0.0371}{0.0282} = 1.3156$$

So, chemical composition (wt%)

$$\text{Fe} = 1 \times 97.95 = 97.95$$

$$\text{C} = 1.0385 \times 0.191 = 0.198$$

$$\text{Si} = 1.1839 \times 0.378 = 0.4475$$

$$\text{Mn} = 1.0578 \times 0.901 = 0.9531$$

$$\text{P} = 1.803 \times 0.0294 = 0.053$$

$$\text{S} = 1.3156 \times 0.0492 = 0.0647$$

Chemical Composition in wt.% in Plain Carbon Steel:

Sample Identification	Fe	C	Si	Mn	P	S
Plain Carbon Steel	97.95	0.198	0.4475	0.9531	0.053	0.0647

Discussion:

1) Why does industry use OES instead of high-accuracy XRF?

Though XRF has higher accuracy than OES, in industry, decisions need to be made quickly. For example, if we delay supplying molten metal from the ladle to the tundish to check its properties before delivery, the metal can lose heat or solidify, which would result in a huge production loss. So, speed is a greater concern in the industry than high accuracy. That's why OES is preferred.

Why can not we conduct analysis using OES in cast iron?

Because in grey cast iron, the flakes are localized, the spark in OES is taken at a single spot. This can lead to high carbon variation at different spots, as the carbon in flakes is highly localized.

Therefore, for any cast iron, chemical analysis is much better than OES.

What is the quantitative analysis problem that can occur in OES?

If certain elements are not detected in OES, it will distribute the values of the detected elements across the rest of the system. By assuming the total as 100%, it will display only the elements present in the detection system, with a slight error due to trace elements.

What does it imply of the chemical composition of my sample?

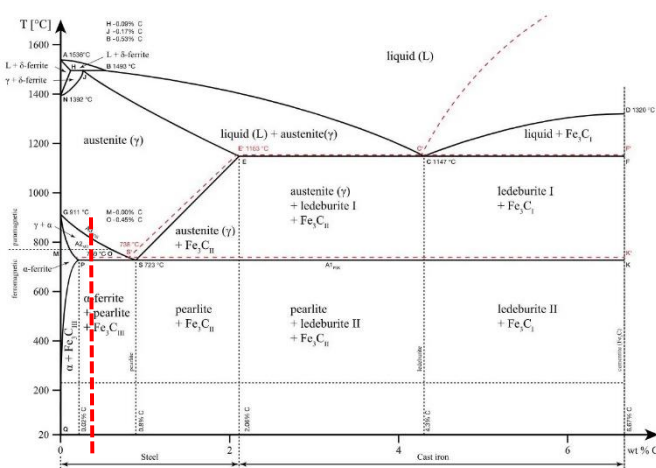


Figure 4: Fe-C diagram

The carbon content of 0.198 wt% indicates the hypoeutectoid steel (less carbon content than eutectoid composition: 0.8%) of the iron-carbon diagram. At room temperature the microstructure will contain proeutectoid ferrite and pearlite.

This low-carbon steel is best suited for structural applications, automotive components, and machinery parts requiring good formability and weldability.

The composition is close to AISI 1018 steel, which contains 0.15–0.20% carbon and 0.60–0.90% manganese. However, the manganese content of 0.9531% is slightly higher, which can improve hardenability. Phosphorus (0.053%) and sulfur (0.0647%) contents are above typical limits and may reduce ductility and toughness, potentially affecting performance during hot working.

The elevated manganese level enhances strength and hardenability, beneficial for applications like case hardening. Silicon content (0.4475%) is slightly above the usual range but can improve strength. High phosphorus and sulfur levels are undesirable as they can lead to brittleness and reduced impact resistance.

It has a near AISI 1018 steel chemistry of 0.20–0.15% carbon and 0.90–0.60% manganese. Manganese is a bit elevated at 0.9531%, and that could yield increased hardenability. Phosphorus is 0.053% and sulfur is 0.0647%, and those are over normal levels and can reduce ductility and toughness and could affect hot working responsiveness.

Grade	C	Si	Mn	P	S
AISI 1018	0.15-0.20	0.10-0.35	0.60-0.90	≤0.04	≤0.05
S355J2G3	≤ 0.22	≤0.55	≤1.6	≤ 0.035	≤ 0.035
For S355J2G3	Cr≤0.3, Mo≤0.08, Ni≤0.3, Cr+Mo+Ni≤0.48, Al≥0.02				

Figure 5: AISI 1018 grade steel

The increased manganese content provides strength and hardenability that can be used in applications like case hardening. Silicon content of 0.4475% is a bit above normal, but can prove to be advantageous to enhance strength. The elevated phosphorus and sulfur content is not desirable because it will cause brittleness and poor impact resistance.